THE LOCAL LIMIT OF SUBLINEAR PREFERENTIAL ATTACHMENT GRAPHS WITH FIXED OUT-DEGREE

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Abstract. We consider the preferential attachment graph in which every new vertex sends $m \geq 2$ edges to the existing vertices, the endpoints being chosen one after another with probability proportional to f(in-degree), where the attachment function f is bounded away from zero, has bounded increments and satisfies $f(k)/k \to 0$. No monotonicity or concavity is assumed. We prove that the total weight of the first n vertices divided by n converges almost surely to the unique solution c_f of an explicit scalar equation, that the empirical in-degree distribution converges almost surely to the law of a pure birth process with rates $mf(k)/c_f$ observed at an independent exponential time, and that the graph converges locally in probability to an infinite random tree. The probability of every finite ball in the limit is given by an explicit integral over the ages of its vertices. This answers, for this class of attachment functions, a question of Deijfen and van der Hofstad on local limits beyond the affine case when the out-degree is fixed.

1. Introduction

1.1. **The model and the question.** Fix an integer $m \geq 2$ and a function $f : \{0, 1, 2, \dots\} \to (0, \infty)$. The graph G_1 consists of the vertex 1 and no edge. Given G_t, whose vertex set is $[t] = \{1, \dots, t\}$, the graph G_{t+1} is obtained by adding the vertex $t + 1$ and m directed edges from $t + 1$ to vertices of $[t]$. The endpoints are chosen one after another. If D_i denotes the current in-degree of $i \in [t]$, which includes the edges already sent by $t + 1$, the next endpoint is i with probability

$$\frac{f(D_i)}{\sum_{j \leq t} f(D_j)}.$$

Thus the in-degrees are updated after every edge, multiple edges may occur, and there are no loops. We write $D_i(n)$ for the in-degree of i in G_n and $S_n = \sum_{i \leq n} f(D_i(n))$ for the total weight.

When $f(k) = k + \delta$ is affine the model is a version of the preferential attachment graph of Barabási and Albert [4]. Bollobás, Riordan, Spencer and Tusnády [8] made that graph precise and determined its degree sequence. In the affine case the weights form a Pólya urn, the graph has an exchangeable representation, and Berger, Borgs, Chayes and Saberi [7] used it to identify the local limit, the Pólya point tree; see [13, 18] and [15, Chapter 5] for extensions, including random out-degrees and additive fitness. Attachment functions that are not affine go back to Krapivsky, Redner and Leyvraz [17] and Krapivsky and Redner [16], who computed the limiting degree distribution of the tree $m = 1$ for attachment proportional to a power a of the degree, which in the in-degree indexing used here is $f(k) = (k + 1)^a$; it has the product form of (1) below. For $m = 1$ the graph is a tree, which embeds in a continuous-time branching process for every f for which that process does not explode and has a Malthusian parameter, and Rudas, Tóth and Valkó [20] obtained its degree distribution and its local limit in this way. Choi, Sethuraman and Venkataramani [9] prove a law of large numbers for the degree counts

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when $f(k)/k \to 0$ in a model in which every step adds one edge, from a new vertex to an old one or between two old vertices, so that the graph has cycles; their proof is a fluid limit, and they remark that it should apply to models in which a bounded number of edges is added at each step. Dereich and Mörters [10, 11] treat concave attachment functions in a model in which a new vertex connects to each old vertex independently, so that the out-degree is random. Open Problem 24 of [2], contributed by Deijfen and van der Hofstad, asks for the local limit when the out-degree is fixed, $m \geq 2$, and f is not affine: the branching process embedding does not extend to this case, and the Pólya urn representation is not available. Banerjee, Deka and Olvera-Cravioto [3] obtain local limits for graphs constructed by collapsing the vertices of a tree with a general attachment function; in a collapsed graph the weight of a vertex is the sum of the weights of the tree vertices it is made of, which is a function of its degree only when f is affine.

1.2. **The result.** We assume

(E1) $f(k) \geq \alpha$ for all k, for some $\alpha > 0$;
(E2) $|f(k+1) - f(k)| \leq L$ for all k, for some $L < \infty$;
(E3) $f(k)/k \to 0$ as $k \to \infty$.

Examples are all functions with values in a compact subset of $(0, \infty)$, the functions $(k+\delta)^a$ with $\delta > 0$ and $0 < a < 1$, and $1 + \log(k+1)$. For $d > 0$ let $p(d) = (p_k(d))_{k \geq 0}$ be the probability vector

$$p_k(d) = \frac{d}{d + mf(k)} \prod_{j=0}^{k-1} \frac{mf(j)}{d + mf(j)}, \tag{1}$$

which is the law of $X(T)$ when X is the pure birth process started at 0 with rates $(m/d)f(k)$ and T is an independent exponential time of mean one.

For a rooted graph (G, u) and $r \geq 0$, let $B_r(G, u)$ be the subgraph induced by the vertices at distance at most r from u, where the distance ignores the orientation of the edges; $B_r(G, u)$ is rooted at u and keeps the orientations. We write $\cong$ for isomorphism of rooted directed multigraphs. A sequence of finite random graphs G_n *converges locally in probability* to a random rooted graph $(\mathcal{T}, o)$ if, for every r and every finite rooted directed multigraph H,

$$\frac{1}{|G_n|} \#\big\{u \in G_n : \ B_r(G_n, u) \cong H\big\} \ \longrightarrow \ \mathbb{P}\big(B_r(\mathcal{T}, o) \cong H\big)$$

in probability [6, 1, 15]; local convergence of directed graphs is set up in [14].

Theorem 1. *Assume* (E1)–(E3).

(i) *There is a unique $c_f > 0$ such that*

$$\sum_{k \geq 1} \prod_{j=0}^{k-1} \frac{mf(j)}{c_f + mf(j)} = m,$$

that is, such that $p(c_f)$ has mean m.

(ii) *Almost surely $S_n/n \to c_f$, and the proportion of vertices of G_n with in-degree k converges to $p_k(c_f)$ for every k, with convergence of the sequences in ℓ^1. Moreover $\max_{i \leq n} D_i(n)/n \to 0$ in probability.*

(iii) *G_n converges locally in probability to a random rooted directed tree $(\mathcal{T}_f, o)$. This tree is infinite and has no multiple edges, every vertex has out-degree m, the in-degree of the root has law $p(c_f)$, and for every admissible ball shape H of radius r the probability $\mathbb{P}(B_r(\mathcal{T}_f, o) \cong H)$ is the integral $\mathbf{P}_H$ given in* (8) *and Proposition 16, with $c = c_f$.*

Part (ii) is the counterpart, for the model with fixed out-degree, of the law of large numbers of [9], and for $m = 1$ the formula for $p(c_f)$ is the degree distribution of [17, 20]; part (iii) answers Open Problem 24 of [2] under (E1)–(E3). Constant f is included and gives uniform attachment, for which $c_f = f(0)$ and $p(c_f)$ is geometric. The theorem does not cover attachment functions with $f(k)/k \to \gamma \in (0, \infty)$, affine or not, functions with unbounded increments, or functions with $\inf f = 0$. Other conventions for the m edges of a vertex, such as in-degrees that are updated only after the m edges, are not treated either.

Remark 2 (The limiting tree)**.** The formula for $\mathbf{P}_H$ has the following reading. Let $\lambda = m/c_f$ and $h_\lambda(u) = \mathbb{E}f(X(u))$, where X is the pure birth process started at 0 with rates $\lambda f(k)$. Every vertex of $\mathcal{T}_f$ has an age in $(0, 1)$, and the age of the root is uniform. A vertex of age x sends its m edges to parents whose ages are independent with density $h_\lambda(\log(x/y))/(c_f x)$ on $(0, x)$; for a vertex reached from one of its parents this applies to the other $m - 1$ edges. The density is uniform only when f is constant. The children of the root, and of any vertex reached from one of its parents, are born at the jump times of a birth process which starts from 0 at the age x of the vertex and jumps from k to $k + 1$ at rate $\lambda f(k)/t$ at time $t \in (x, 1]$. For a vertex of age y reached from one of its children, of age x, the law of this birth process is weighted by f evaluated at its state just before time x, and a jump is added at time x. For $f \equiv 1$ this is the local limit of uniform attachment.

1.3. **The proof.** Without an urn representation the first difficulty is the normalisation S_n. We write the empirical in-degree distribution as a stochastic approximation in ℓ^1, in the sense of [5, 19], and identify its limit through a time change that turns the limiting differential equation into a linear one. The same time change appears in [9], where the linear equation is treated by semigroup methods; here it is solved by a renewal equation, and the key renewal theorem gives convergence to $p(c_f)$ uniformly over the compact set of probability vectors with mean at most m (Section 3). No monotonicity of f is needed, and the equilibrium need not be stable in any linearised sense that we would have to verify. The local limit is then obtained from an exact change of measure: the probability that prescribed slots choose prescribed endpoints equals an expectation, for the process in which these choices are forced, of a product of ratios $f(D)/S$ (Section 4). Finitely many forced choices do not change the normalisation, the in-degrees of finitely many tagged vertices become independent birth processes in the logarithm of the age, and a count of embeddings with a second moment computation for two independent roots completes the proof. The a priori bounds on the size of balls and on old vertices are in Section 2. Sections 2–4 are written for bounded f; Section 5 gives the changes needed under (E1)–(E3), the main one being a moment bound for the in-degree of a vertex which replaces the uniform bound on the probability of an edge. Section 6 compares the formulas with simulations.

2. A priori bounds

Throughout this section f satisfies $0 < \alpha \le f(k) \le \beta < \infty$ for all $k \ge 0$, and $C = \beta/\alpha$. The m edges sent by a vertex t are called the *slots* of t and are numbered $1, \dots, m$ in the order in which their endpoints are chosen.

Lemma 3 (Joint edge bound)**.** *Let E be a finite multiset of pairs $\{i, j\}$ of distinct labels in $[n]$, with e elements counted with multiplicity. The probability that G_n contains every pair of E as an edge, with at least the prescribed multiplicity, is at most*

$$\prod_{\{i,j\}\in E} \frac{mC}{\max(i,j) - 1} \;\le\; (2mC)^e \prod_{\{i,j\}\in E} \frac{1}{\max(i,j)}.$$

Proof. At any slot of a vertex t and for any history, a given vertex $i < t$ is chosen with conditional probability $f(D_i)/S \le \beta/(\alpha(t-1)) = C/(t-1)$, because the denominator is a sum of $t-1$ weights, each at least α. Assign to every element $\{i,j\}$ of E a slot of its younger endpoint $\max(i,j)$, distinct elements receiving distinct slots. Exposing the slots in chronological order and conditioning successively on the past, the probability that every assigned slot chooses its prescribed endpoint is at most the product of $C/(\max(i,j)-1)$ over E. There are at most m^e assignments, and the event in the statement is the union of the corresponding events. The second inequality uses $t-1 \ge t/2$ for $t \ge 2$. □

Lemma 4. *The symmetric $n \times n$ matrix $H = (1/\max(i,j))_{i,j \le n}$ has operator norm at most 4 on ℓ^2.*

Proof. Let $w_i = i^{-1/2}$. Then $\sum_{j \le i} H_{ij} w_j = i^{-1} \sum_{j \le i} j^{-1/2} \le 2i^{-1/2}$ and $\sum_{j>i} H_{ij} w_j = \sum_{j>i} j^{-3/2} \le 2i^{-1/2}$, so $Hw \le 4w$ coordinatewise. For real x, the inequality $2|x_i x_j| \le x_i^2 w_j/w_i + x_j^2 w_i/w_j$ and the symmetry of H give $|x^{\mathsf T} H x| \le \sum_i x_i^2 (Hw)_i/w_i \le 4\|x\|^2$. □

Let U_n be uniform on $[n]$ and independent of G_n, and let $B_r(G_n, U_n)$ be the ball of radius r around U_n for the graph distance that ignores orientation. Put $B = 8mC$.

Proposition 5 (Balls are bounded in mean and avoid old vertices)**.** *For all $n \ge 1$, $r \ge 0$ and $1 \le K \le n$,*

$$\mathbb{E}\,|B_r(G_n,U_n)| \le \sum_{l=0}^{r} B^l, \qquad \mathbb{P}\big(B_r(G_n,U_n) \cap [K] \ne \emptyset\big) \le \Big(\sum_{l=0}^{r} B^l\Big)\sqrt{K/n}.$$

Proof. A vertex at distance l from U_n is the endpoint of a path $v_0 = U_n, v_1, \dots, v_l$ with distinct vertices, hence distinct edges. By Lemma 3, the expected number of such paths ending in a set A is at most

$$\frac{(2mC)^l}{n} \sum_{v_0,\dots,v_l} \prod_{k=1}^{l} \frac{1}{\max(v_{k-1},v_k)} \mathbf{1}_{v_l \in A} = \frac{(2mC)^l}{n}\, \mathbf{1}^{\mathsf T} H^l \mathbf{1}_A \le \frac{(2mC)^l}{n}\, \sqrt{n}\, 4^l \sqrt{|A|},$$

by Lemma 4. Take $A = [n]$ and $A = [K]$ and sum over $l \le r$. □

Proposition 6 (No short cycles)**.** *For every fixed r, the probability that $B_r(G_n, U_n)$ contains a cycle or a multiple edge tends to 0.*

Proof. Fix $\varepsilon \in (0,1)$ and let $K = \lfloor \varepsilon n \rfloor$. Suppose that the ball avoids $[K]$ and is not a simple tree. Take a breadth-first spanning tree of the ball and an edge of the ball outside it (a second copy of a multiple edge counts as such an edge). This edge together with the tree paths from its endpoints to the root forms a connected rooted multigraph with $v \le 2r+1$ vertices and exactly v edges, all of whose labels exceed K. There are finitely many such rooted shapes. For a fixed shape there are at most n^v labellings; the root label equals U_n with probability $1/n$; and by Lemma 3 the v edges are present with probability at most $(2mC/(\varepsilon n))^v$. The union bound gives $O_{r,\varepsilon}(1/n)$. With Proposition 5,

$$\limsup_{n \to \infty} \mathbb{P}\big(B_r(G_n,U_n) \text{ is not a simple tree}\big) \le \Big(\sum_{l \le r} B^l\Big)\sqrt{\varepsilon},$$

and ε is arbitrary. □

Corollary 7. *The laws of the rooted graphs (G_n, U_n) are tight for the local topology. Every subsequential limit is almost surely a locally finite simple tree in which every vertex has out-degree m; in particular it is infinite.*

Proof. A vertex has at most m outgoing edges, so there are finitely many rooted directed multigraphs with at most M vertices that can occur as balls. Tightness follows from the first bound of Proposition 5 and Markov's inequality, applied for each r with a threshold M_r chosen so that the exceptional probabilities are summable. Limits are simple trees by Proposition 6. Every vertex other than vertex 1 has out-degree m, all edges leaving B_r are seen in B_{r+1}, and B_{r+1} contains vertex 1 with probability at most $(\sum_{l\le r+1} B^l)n^{-1/2}$; hence all out-degrees in a limit equal m. A finite tree with v vertices has $v-1<mv$ edges, so the limit is infinite. □

Lemma 8 (Degrees)**.** *Let $D_i(n)$ be the in-degree of i in G_n. Then $\mathbb{E}D_{U_n}(n)^2 \le mC+2(mC)^2$, and for every $\delta>0$,*

$$\mathbb{P}\Big(\max_{i\le n} D_i(n)\ge \delta n\Big) \le \frac{mC+2(mC)^2}{\delta^2 n}.$$

Proof. Let $L_i = mC\sum_{s=i}^{n-1} 1/s$. Summing the single-slot bound $C/(t-1)$ over the slots of $t=i+1,\dots,n$ gives $\mathbb{E}D_i(n)\le L_i$, and summing the product of two such bounds over ordered pairs of distinct slots gives $\mathbb{E}[D_i(n)(D_i(n)-1)]\le L_i^2$. Exchanging the order of summation,

$$\sum_{i\le n} L_i = mC(n-1), \qquad \sum_{i\le n} L_i^2 = (mC)^2 \sum_{s,t<n} \frac{\min(s,t)}{st} \le 2(mC)^2(n-1).$$

Hence $\sum_i \mathbb{E}D_i(n)^2 \le n\,(mC+2(mC)^2)$, which gives the first claim, and the second follows from Markov's inequality and a union bound. □

The second statement says that no vertex receives a positive fraction of the edges. The first gives uniform integrability of the degree of the root, so the root of every subsequential limit has mean in-degree m.

3. The normalisation and the degree law

In this section $0<\alpha\le f\le\beta<\infty$. Let $N_k(n)$ be the number of vertices of G_n with in-degree k, $y_n=(N_k(n)/n)_{k\ge0}$, and $S_n=\sum_{i\le n} f(D_i(n)) = n\,s(y_n)$, where $s(y)=\sum_k f(k)y_k$.

3.1. The scalar equation.

For $d>0$ let

$$R_k(d)=\prod_{j=0}^{k-1}\frac{mf(j)}{d+mf(j)} \quad (k\ge1), \qquad M(d)=\sum_{k\ge1} R_k(d),$$

and let $p(d)=(p_k(d))_{k\ge0}$ be the law of $X(T)$, where X is the pure birth process started at 0 that jumps from k to $k+1$ at rate $(m/d)f(k)$, and T is an independent exponential time of mean one.

Lemma 9. (i) $\mathbb{P}(X(T)\ge k)=R_k(d)$, *so $p(d)$ has mean $M(d)$, and*

$$p_0(d)=\frac{d}{d+mf(0)}, \qquad d\,p_k(d)=mf(k-1)p_{k-1}(d)-mf(k)p_k(d) \quad (k\ge1). \tag{2}$$

(ii) $d\,M(d)=m\sum_{k\ge0} f(k)p_k(d)$. (iii) *There is exactly one $c_f>0$ with $M(c_f)=m$, and $c_f\in[\alpha,\beta]$.*

Proof. (i) From state j the process jumps before the killing time with probability $mf(j)/(d+mf(j))$, and (2) is the same computation written for the point masses. (ii) $R_k(d)\le(m\beta/(d+m\beta))^k$, so all sums converge absolutely; multiply (2) by k and sum over $k\ge1$. (iii) Each R_k is continuous and strictly decreasing in d, the series converges uniformly on $[d_0,\infty)$ for every $d_0>0$, $M(d)\le m\beta/d\to0$ as $d\to\infty$, and $M(d)\ge\sum_{k\le K}R_k(d)\to K$ as $d\to0$ for every K.

At $d = \alpha$ every factor is at least $m/(m+1)$, so $M(\alpha) \geq m$; at $d = \beta$ every factor is at most $m/(m+1)$, so $M(\beta) \leq m$. □

Theorem 10. *Almost surely $y_n \to p(c_f)$ in ℓ^1 and $S_n/n \to c_f$.*

The rest of the section proves Theorem 10. No monotonicity of f is used.

3.2. **Stochastic approximation in ℓ^1.** Let e_0 be the unit mass at 0 and let Q be the bounded operator on ℓ^1 given by

$$(Qy)_0 = -f(0)y_0, \qquad (Qy)_k = f(k-1)y_{k-1} - f(k)y_k \quad (k \geq 1),$$

the generator of the pure birth process with rates $f(k)$ acting on measures; $\|Q\| \leq 2\beta$ and $\sum_k (Qy)_k = 0$. On probability vectors put

$$F(y) = e_0 + \frac{m}{s(y)}\, Qy - y.$$

Since $s \geq \alpha$ and $|s(y) - s(z)| \leq \beta \|y - z\|_1$, F is Lipschitz in ℓ^1 on probability vectors.

At a slot of vertex $n+1$, if the current counts are N and the current total weight is S, a vertex of in-degree k is chosen with probability $f(k)N_k/S$, so the conditional mean increment of the count vector is QN/S. During the m slots N changes by at most $2m$ in ℓ^1 and S by at most $2m\beta$, while $S \geq \alpha n$ and $\|QN\|_1 \leq 2\beta n$; hence QN/S differs from $Qy_n/s(y_n)$ by $O(1/n)$ in ℓ^1, uniformly in the configuration. After the m slots the new vertex adds e_0. Therefore

$$y_{n+1} = y_n + \frac{1}{n+1}\big(F(y_n) + \eta_{n+1} + \rho_n\big), \tag{3}$$

where $\mathbb{E}[\eta_{n+1} \mid \mathcal{F}_n] = 0$ coordinatewise, $\|\eta_{n+1}\|_1 \leq 4m + 2$ and $\|\rho_n\|_1 \leq K_0/n$ for a constant K_0.

Every y_n lies in

$$\mathcal{K} = \Big\{ y \geq 0 : \ \sum_k y_k = 1, \ \sum_k k\, y_k \leq m \Big\},$$

because the total in-degree of G_n is $m(n-1)$. The set $\mathcal{K}$ is compact in ℓ^1: the mass above J is at most $m/(J+1)$ uniformly on $\mathcal{K}$, and the mean is lower semicontinuous. For $y \in \mathcal{K}$ and $J \geq 1$,

$$\|F(y)\|_1 \leq 2 + \frac{2m\beta}{\alpha}, \qquad \sum_{k>J} |F(y)_k| \leq \Big(1 + \frac{2m\beta}{\alpha}\Big)\frac{m}{J}. \tag{4}$$

Lemma 11 (Noise)**.** *Let $\tau_n = \sum_{j<n} 1/(j+1)$. Almost surely, for every $T < \infty$,*

$$\lim_{N\to\infty} \sup_{j \geq N:\ \tau_j - \tau_N \leq T} \Big\| \sum_{n=N}^{j-1} \frac{\eta_{n+1}}{n+1} \Big\|_1 = 0.$$

Proof. For each coordinate k the series $\sum_n \eta_{n+1,k}/(n+1)$ is a martingale with summable increment variances, so it converges almost surely; work on the event where this holds for all k. The coordinates $k \leq J$ of the partial sums in the statement then tend to 0 uniformly in j. For the coordinates $k > J$ use (3) summed from N to $j-1$: the noise sum equals $y_j - y_N$ minus the sums of the F terms and of the ρ terms. The mass of y_j and y_N above J is at most $m/(J+1)$ each, the F terms contribute at most a constant times $(T+1)/J$ by (4), and the ρ terms at most $K_0 \sum_{n \geq N} 1/(n(n+1))$. Hence the lim sup in N of the supremum is at most a constant times $(T+1)/J$, and J is arbitrary. □

The argument avoids any convergence theorem for ℓ^1-valued martingales: the tails are controlled by the recursion itself.

3.3. **The limiting flow.** Let $P_t = e^{tQ}$ be the semigroup of the pure birth process with rates $f(k)$; it is positive, preserves mass, and $\|P_t\| = 1$ on ℓ^1. Let

$$A = Q + \frac{1}{m}\, e_0\, s(\cdot), \qquad w(t) = e^{tA} y,$$

for a probability vector y. The operator A is bounded and e^{tA} is positive.

Lemma 12. *Let $|w| = \sum_k w_k$ and $z(t) = w(t)/|w(t)|$. Then $z' = \frac{s(z)}{m} F(z)$. Consequently the solution of $\dot y = F(y)$ started at a probability vector y is $\phi_\tau(y) = z(t(\tau))$, where $\mathrm{d}\tau/\mathrm{d}t = s(z(t))/m \in [\alpha/m, \beta/m]$; it exists for all times, stays in the probability vectors, and $\phi_\tau(\mathcal{K}) \subset \mathcal{K}$.*

Proof. Since Q preserves mass, $|w|' = s(w)/m$, and therefore $z' = Az - \frac{s(z)}{m} z = Qz + \frac{s(z)}{m}(e_0 - z) = \frac{s(z)}{m} F(z)$. Uniqueness for $\dot y = F(y)$ follows from the Lipschitz property. For y with finite mean, $\sum_k k\,(Qy)_k = s(y)$, so the mean $\mu(\tau)$ of $\phi_\tau(y)$ satisfies $\mu' = m - \mu$, and $\mu(0) \le m$ implies $\mu(\tau) \le m$. (The operator Q is bounded for the weighted norm $\sum_k (1+k)|y_k|$, so the mean stays finite and the computation is legitimate.) □

Proposition 13 (Uniform attraction)**.** $\sup_{y \in \mathcal{K}} \|\phi_\tau(y) - p(c_f)\|_1 \to 0$ *as* $\tau \to \infty$.

Proof. Let $g_y(t) = s(P_t y)$ and $h(t) = s(P_t e_0)$; both take values in $[\alpha, \beta]$ and are Lipschitz. Variation of constants gives

$$w(t) = P_t y + \frac{1}{m} \int_0^t P_{t-u} e_0\, \sigma(u)\, \mathrm{d}u, \qquad \sigma(t) = g_y(t) + \frac{1}{m} \int_0^t h(t-u)\, \sigma(u)\, \mathrm{d}u, \tag{5}$$

with $\sigma(t) = s(w(t))$. The function $\gamma \mapsto \frac{1}{m} \int_0^\infty e^{-\gamma t} h(t)\, \mathrm{d}t$ is continuous and strictly decreasing, and lies between $\alpha/(m\gamma)$ and $\beta/(m\gamma)$. Let $\gamma > 0$ be the value at which it equals one; then $\gamma \in [\alpha/m, \beta/m]$, and $\kappa(t) = e^{-\gamma t} h(t)/m$ is a continuous, positive probability density on $[0, \infty)$ with finite mean $\bar\mu$, bounded by $(\beta/m) e^{-\gamma t}$, hence directly Riemann integrable. Let $v = \sum_{j \ge 1} \kappa^{*j}$ be the renewal density; the series converges locally uniformly since $\kappa^{*j}(t) \le (\beta/m)^j t^{j-1}/(j-1)!$. The key renewal theorem [12, Chapter XI], applied to the function κ itself, gives $v(t) \to 1/\bar\mu$ as $t \to \infty$; in particular v is bounded.

Multiplying the second equation of (5) by $e^{-\gamma t}$ gives a renewal equation for $\tilde\sigma(t) = e^{-\gamma t}\sigma(t)$ with $\tilde g_y(t) = e^{-\gamma t} g_y(t)$, whose solution is

$$\tilde\sigma(t) = \tilde g_y(t) + \int_0^t v(t-u)\, \tilde g_y(u)\, \mathrm{d}u.$$

Since $\alpha e^{-\gamma u} \le \tilde g_y(u) \le \beta e^{-\gamma u}$ for every probability vector y, dominated convergence gives

$$\tilde\sigma(t) \to C_y := \frac{1}{\bar\mu} \int_0^\infty \tilde g_y(u)\, \mathrm{d}u \in \Big[\frac{\alpha}{\gamma\bar\mu}, \frac{\beta}{\gamma\bar\mu}\Big]$$

uniformly in y. Multiplying the first equation of (5) by $e^{-\gamma t}$ and using $\|P_u e_0\|_1 = 1$, we get $e^{-\gamma t} w(t) \to \frac{C_y}{m}\, \mathcal{R}$ in ℓ^1, uniformly in y, where $\mathcal{R} = \int_0^\infty e^{-\gamma u} P_u e_0\, \mathrm{d}u$ has mass $1/\gamma$. Hence $z(t) \to \pi := \gamma\mathcal{R}$ uniformly in y, and by Lemma 12 (the clock $\tau(t)$ runs at speed at most β/m, so $t(\tau) \ge m\tau/\beta$) $\phi_\tau(y) \to \pi$ uniformly in y.

The vector π is the law of the pure birth process with rates $f(k)$ at an independent exponential time of rate γ, that is, $\pi = p(m\gamma)$. Its mean is $\int_0^\infty \gamma e^{-\gamma t} \int_0^t h(u)\, \mathrm{d}u\, \mathrm{d}t = \int_0^\infty e^{-\gamma u} h(u)\, \mathrm{d}u = m$ by the choice of γ. Thus $M(m\gamma) = m$, and Lemma 9(iii) gives $m\gamma = c_f$. □

Proof of Theorem 10. Interpolate y_n linearly at the times τ_n and call the path $y(\cdot)$. By (3), Lemma 11, the bound $\|\rho_n\|_1 \le K_0/n$, the Lipschitz property of F and Gronwall's inequality,

almost surely for every T

$$\sup_{0\le u\le T}\big\|y(\tau+u)-\phi_u(y(\tau))\big\|_1\to 0 \qquad (\tau\to\infty);$$

the replacement of the step sums of $F(y_n)$ by an integral costs $O(\sum_{n\ge N}(n+1)^{-2})$ because $\|y_{n+1}-y_n\|_1=O(1/n)$. Given $\varepsilon>0$ choose T with $\sup_{y\in\mathcal{K}}\|\phi_T(y)-p(c_f)\|_1<\varepsilon$ (Proposition 13); the interpolated path stays in the convex set $\mathcal{K}$, so $\limsup_\tau\|y(\tau)-p(c_f)\|_1\le\varepsilon$ almost surely. Finally $S_n/n=s(y_n)\to s(p(c_f))=c_f$ by Lemma 9(ii), since s is continuous on ℓ^1. □

Example 14 (Two levels)**.** Let $f(0)=a$ and $f(k)=b$ for $k\ge 1$. Then $S_n=bn+(a-b)N_0(n)$, and $z=p_0(c_f)$ is the root in $(0,1)$ of

$$(a-b)z^2+(2b+ma-a)z-b=0, \qquad c_f=b+(a-b)z,$$

with $p_1=\lambda az/(1+\lambda b)$ and $p_k=p_1\,(\lambda b/(1+\lambda b))^{k-1}$, $\lambda=m/c_f$. For $m=2$, $a=1$, $b=2$ this gives $z=(5-\sqrt{17})/2$ and $c_f=(\sqrt{17}-1)/2$. For $a=b$ the law is geometric with ratio $m/(m+1)$, the degree law of uniform attachment.

4. The local limit

In this section $0<\alpha\le f\le\beta<\infty$, $c=c_f$ and $\lambda=m/c$. By Theorem 10, $S_t/t\to c$ almost surely; since $\alpha\le S_t/t\le\beta$, for every $\varepsilon>0$

$$\mathbb{E}\sup_{t\ge\varepsilon n}\big|S_t/t-c\big|\to 0 \qquad (n\to\infty). \tag{6}$$

4.1. **Ball shapes and the limiting probabilities.** Fix $r\ge 1$. A *ball shape* is a finite rooted directed simple tree H with vertex set V and root o in which every vertex is within distance r of o and every vertex at distance r is a leaf. Let I be the set of vertices at distance less than r (the interior), and for $j\in V$ let q_j and k_j be the out-degree and the in-degree of j in H. We write $j\to i$ for an edge sent by j and received by i. The shape is *admissible* if $q_j=m$ for every $j\in I$ and $q_j\le m$ for every j. Let $A(H)$ be the number of automorphisms of H that fix the root and preserve orientations, and $(m)_q=m!/(m-q)!$.

An *age vector* is $x\in(0,1)^V$ with distinct coordinates; it is *compatible* if $x_j>x_i$ for every edge $j\to i$. Given a compatible age vector, let $(Y_i)_{i\in V}$ be independent processes, where Y_i is defined on $[x_i,1]$, starts at 0, jumps by $+1$ at rate $\lambda f(Y_i(t))/t$ at time t, and in addition jumps by $+1$ at each of the times x_j, $j\to i$. Define

$$\Phi_H(x)=\prod_{j\in V}(m)_{q_j}\ \prod_{j\to i}\frac{1}{c\,x_j}\ \mathbb{E}\Big[\prod_{j\to i}f\big(Y_i(x_j-)\big)\prod_{i\in I}\mathbf{1}_{\{Y_i(1)=k_i\}}\Big] \tag{7}$$

for compatible x and $\Phi_H(x)=0$ otherwise, and

$$\mathbf{P}_H=\frac{1}{A(H)}\int_{(0,1)^V}\Phi_H(x)\,\mathrm{d}x. \tag{8}$$

Theorem 15. *For every $r\ge 1$ and every admissible ball shape H of radius r,*

$$\frac{1}{n}\,\#\big\{u\in[n]:\ B_r(G_n,u)\cong H\big\}\ \longrightarrow\ \mathbf{P}_H \qquad \textit{in probability,}$$

and the proportion of u whose ball $B_r(G_n,u)$ is not an admissible ball shape tends to 0 in probability. Consequently G_n converges locally in probability to the random rooted tree whose balls have the laws $(\mathbf{P}_H)$; this tree is infinite, simple, and all its vertices have out-degree m.

The expectation in (7) is a product of explicit factors.

Proposition 16. *Let X be the pure birth process started at 0 with rates $\lambda f(k)$, and $h_\lambda(u) = \mathbb{E}f(X(u))$. For compatible x the expectation in (7) equals $\prod_{i\in V} F_i$, where: for $i \in I$ with children of ages $t_1 < \cdots < t_k$ ($k = k_i$), and $t_0 = x_i$, $t_{k+1} = 1$,*

$$F_i = \prod_{\ell=0}^{k-1} f(\ell) \prod_{\ell=0}^{k} \Big(\frac{t_\ell}{t_{\ell+1}}\Big)^{\lambda f(\ell)};$$

for $i \notin I$, $F_i = 1$ if $k_i = 0$ and $F_i = h_\lambda(\log(t_1/x_i))$ if $k_i = 1$, where t_1 is the age of the child of i.

Proof. The processes are independent, so the expectation factorises. For $i \in I$ the event $Y_i(1) = k_i$ says that Y_i makes no jump other than the k_i forced ones; on this event $Y_i = \ell$ on $[t_\ell, t_{\ell+1})$, the probability of no other jump on that interval is $\exp(-\lambda f(\ell)\int_{t_\ell}^{t_{\ell+1}} \mathrm{d}t/t)$, and the weight at the ℓth forced jump is $f(\ell-1)$. A vertex $i \notin I$ is a leaf of H with no constraint at time 1; if its neighbour is its child of age t_1, then $Y_i(t_1-)$ has the law of $X(\log(t_1/x_i))$. □

For instance, if H is the star in which the root has no child, the root contributes $x_o^{\lambda f(0)}$, and each of its m parents, of age $y < x_o$, contributes $h_\lambda(\log(x_o/y))/(c\,x_o)$, which integrates to $\int_0^\infty e^{-u}h_\lambda(u)\,\mathrm{d}u/c = 1$ over $y \in (0, x_o)$ by Lemma 9(ii); since $(m)_m = A(H) = m!$, this gives $\mathbf{P}_H = \int_0^1 x^{\lambda f(0)}\,\mathrm{d}x = p_0(c)$, in agreement with Theorem 10. The same computation shows that the age y of a parent of a vertex of age x has density $h_\lambda(\log(x/y))/(cx)$ on $(0, x)$, which is not uniform unless f is constant.

4.2. **Forced slots.** Fix an admissible H with $e = |V| - 1$ edges, an injective labelling $\iota : V \to [n]$ with $\iota(j) > \iota(i)$ for every edge $j \to i$, and a *slot assignment* σ, which gives to every edge $j \to i$ a slot of $\iota(j)$, different edges receiving different slots. There are $\prod_j (m)_{q_j}$ slot assignments. Let $E_{\iota,\sigma}$ be the event that for every edge $j \to i$ the assigned slot of $\iota(j)$ chooses $\iota(i)$, and let $D_\iota = \{D_{\iota(i)}(n) = k_i \text{ for all } i \in I\}$.

The *forced process* is the graph process in which every assigned slot chooses its prescribed endpoint with probability one and all other slots follow the original rule. Write $\mathbb{P}^\sigma$, $\mathbb{E}^\sigma$ for its law. Multiplying the transition probabilities along a history gives, for every nonnegative function Ψ of the history up to time n,

$$\mathbb{E}\big[\Psi\,\mathbf{1}_{E_{\iota,\sigma}}\big] = \mathbb{E}^\sigma\Big[\Psi \prod_{j\to i} \frac{f(D^-_{j\to i})}{S^-_{j\to i}}\Big], \tag{9}$$

where $D^-_{j\to i}$ is the in-degree of $\iota(i)$ and $S^-_{j\to i}$ the total weight of the eligible vertices immediately before the slot assigned to $j \to i$.

Lemma 17 (The forced process has the same normalisation)**.** *Suppose that all labels $\iota(j)$ are at least εn. Under $\mathbb{P}^\sigma$, $\sup_{\varepsilon n \le t \le n} |S_t/t - c| \to 0$ in L^1 as $n \to \infty$, uniformly in ι and σ.*

Proof. Couple the forced process with the original one: they coincide until the first assigned slot, and afterwards, at each slot that is not assigned, the two endpoints are chosen by a maximal coupling of the two conditional laws. Let $\bar K$ be the number of labels at which the two in-degree sequences have differed at some time. If the two weight vectors at a slot of vertex $t+1$ are w and w' with totals $W, W' \ge \alpha t$, then

$$\big\|w/W - w'/W'\big\|_1 \le \frac{2\|w - w'\|_1}{W} \le \frac{2\beta\bar K}{\alpha t},$$

so the two choices differ with probability at most $C\bar K/t$, in which case $\bar K$ increases by at most 2; at an assigned slot $\bar K$ increases by at most 2. Hence $\mathbb{E}\bar K \le 2e\exp(2mC\sum_{t\ge\varepsilon n/2}^{n} 1/t) \le K_\varepsilon$,

a constant that does not depend on n. Since the two values of S_t differ by at most $\beta\bar{K}$, the claim follows from (6). □

Lemma 18 (Tagged degrees)**.** *Let $\iota = \iota_n$ and σ be as above, and suppose that $\iota_n(j)/n \to x_j$ for all j, where x is a compatible age vector with all $x_j \geq \varepsilon$. Under $\mathbb{P}^\sigma$, the vector formed by the in-degrees $D^-_{j\to i}$ $(j \to i)$ and $D_{\iota(i)}(n)$ $(i \in V)$ converges in total variation to the vector formed by $Y_i(x_j-)$ and $Y_i(1)$.*

Proof. The vertices $\iota(i)$, $i \in V$, are called tagged. Consider a slot of a vertex $t+1$ that is not assigned. Given the history, the tagged vertex $\iota(i) \leq t$ is chosen with probability $f(D_{\iota(i)})/S$, where S is the current total weight, and at most one tagged vertex is chosen. Compare with independent Bernoulli variables of parameters $f(D_{\iota(i)})/(ct)$: the total variation distance between the two laws of the increment of the tagged vector is at most

$$K_1 \, \frac{|S/t - c|}{t} + \frac{K_1}{t^2},$$

with K_1 depending on $|V|, m, \alpha, \beta$ only, because $S \geq \alpha t$, $c \geq \alpha$ and the probability that two of the Bernoulli variables equal one is $O(t^{-2})$. At an assigned slot the prescribed tagged vertex gains one and the others gain nothing, in both systems. Using maximal couplings slot by slot, the tagged vector of the forced process coincides up to time n with a system of independent chains, one for each tagged vertex, which start at 0 at the birth of the vertex, jump at each unassigned slot of a vertex $t+1$ with probability $f(\text{state})/(ct)$, and jump at their assigned slots, except on an event of probability at most

$$K_1 m \log(1/\varepsilon) \; \mathbb{E}^\sigma \sup_{t \geq \varepsilon n} |S_t/t - c| + O\big(1/(\varepsilon n)\big),$$

which tends to 0 by Lemma 17 (within a batch S differs from S_t by at most $2m\beta$).

It remains to compare each independent chain with Y_i. Over the slots of vertex $t+1$ the chain in state k jumps exactly once with probability $mf(k)/(ct) + O(t^{-2})$ and more than once with probability $O(t^{-2})$; the process Y_i in state k has the same two estimates on the age interval $[t/n, (t+1)/n]$, since $\int \lambda f(k) \, \mathrm{d}s/s = \lambda f(k) \log(1 + 1/t)$. The total variation distance between the two paths, observed at the ages t/n, is therefore $O(\sum_{t \geq \varepsilon n} t^{-2}) = O(1/(\varepsilon n))$, apart from the effect of the forced times, which are $\iota_n(j)/n$ for the chain and x_j for Y_i: moving a forced time by δ changes the law of the observed vector by at most the probability of an unforced jump in a window of length δ, which is $O(\delta/\varepsilon)$. Finally the in-degree of a target immediately before its assigned slot differs from its in-degree at the beginning of the batch with probability $O(1/n)$. □

Proposition 19. *Under the assumptions of Lemma 18,*

$$n^e \, \mathbb{P}\big(E_{\iota_n,\sigma} \cap D_{\iota_n}\big) \;\longrightarrow\; \prod_{j \to i} \frac{1}{c \, x_j} \; \mathbb{E}\Big[\prod_{j \to i} f\big(Y_i(x_j-)\big) \prod_{i \in I} \mathbf{1}_{\{Y_i(1) = k_i\}}\Big],$$

and the left-hand side is bounded by $(4C/\varepsilon)^e$ for all labellings with labels at least εn and n large. The convergence is uniform on every compact set of compatible age vectors.

Proof. Apply (9) with $\Psi = \mathbf{1}_{D_\iota}$. Each factor $nf(D^-)/S^-$ is at most $n\beta/(\alpha(\varepsilon n - 1))$, and $S^-_{j\to i}/n \to c \, x_j$ in L^1 under $\mathbb{P}^\sigma$ by Lemma 17. The numerators and the indicator are functions of the vector of Lemma 18. The limit is continuous in x on the set of compatible age vectors, and convergence along every sequence $\iota_n(\cdot)/n \to x$ gives the uniformity. □

4.3. **Proof of Theorem 15.** *Counting.* Let $\mathcal{A}_\iota$ be the event that ι is an isomorphism from H onto $B_r(G_n, \iota(o))$. A ball isomorphic to H has exactly $A(H)$ such isomorphisms, so

$$\mathbb{P}\big(B_r(G_n, U_n) \cong H\big) = \frac{1}{n\,A(H)} \sum_\iota \mathbb{P}(\mathcal{A}_\iota). \tag{10}$$

We compare $\mathbf{1}_{\mathcal{A}_\iota}$ with $\sum_\sigma \mathbf{1}_{E_{\iota,\sigma}\cap D_\iota}$. Every edge of H has an endpoint in I. On D_ι no edge $j \to i$ with $i \in I$ can be doubled, because the in-degree of $\iota(i)$ is exactly the number of its children; and an edge $j \to i$ with $j \in I$ cannot be doubled on $E_{\iota,\sigma}$, because all m slots of $\iota(j)$ are assigned to distinct endpoints. Hence at most one σ contributes, and on $\mathcal{A}_\iota$ exactly one does. Conversely, on $E_{\iota,\sigma} \cap D_\iota$ all edges of G_n at the vertices $\iota(i)$, $i \in I$, are images of edges of H; by induction on the distance, the vertices at distance $d \le r$ from $\iota(o)$ in G_n are the images of the vertices at distance d in H. Thus $\mathcal{A}_\iota$ can fail only if G_n has an edge between the images of two vertices at distance r. If all labels are at least εn, Lemma 3 bounds the probability of $E_{\iota,\sigma}$ together with one such extra edge by $(4mC/(\varepsilon n))^{e+1}$. There are at most $n^{|V|} = n^{e+1}$ labellings, so

$$\mathbb{P}\big(B_r(G_n, U_n) \cong H,\ B_r(G_n, U_n) \subset [\varepsilon n, n]\big) = \frac{1}{A(H)}\ \frac{1}{n^{|V|}} \sum_{\iota \ge \varepsilon n} \sum_\sigma n^e\, \mathbb{P}(E_{\iota,\sigma} \cap D_\iota) + O_\varepsilon(1/n). \tag{11}$$

Riemann sums. By Proposition 19 the summands are bounded, and they converge uniformly on compact sets of compatible age vectors; the limit does not depend on σ. Labellings for which two ages are within δ form a proportion $O(\delta + 1/n)$ of all labellings. Letting $n \to \infty$ and then $\delta \to 0$, the right-hand side of (11) converges to $A(H)^{-1} \int_{[\varepsilon,1]^V} \Phi_H(x)\,\mathrm{d}x$.

The cutoff. By Proposition 5 the left-hand side of (11) differs from $\mathbb{P}(B_r(G_n, U_n) \cong H)$ by at most $(\sum_{l \le r} B^l)\sqrt{\varepsilon}$. The integrals over $[\varepsilon, 1]^V$ increase as ε decreases and are bounded by one, so Φ_H is integrable and $\mathbb{P}(B_r(G_n, U_n) \cong H) \to \mathbf{P}_H$. Balls that are not admissible ball shapes have vanishing probability by Proposition 6 and Corollary 7, and tightness shows that $(\mathbf{P}_H)_H$ is a probability distribution for each r.

Two roots. Let U_n, U_n' be independent uniform vertices. Their balls of radius r intersect with probability at most $\mathbb{E}|B_{2r}(G_n, U_n)|/n = O(1/n)$. For two admissible shapes H_1, H_2 repeat the argument with the disjoint union $H_1 \sqcup H_2$, which has $|V_1| + |V_2| - 2$ edges, and two root factors $1/n^2$. An edge between the image of H_1 and the image of H_2 can only join vertices at distance r from the two roots and does not change either ball, so the counting step is unchanged. No forced jump links the two components, the processes Y_i are independent, and the limiting density is $\Phi_{H_1}\Phi_{H_2}$. Therefore

$$\mathbb{P}\big(B_r(G_n, U_n) \cong H_1,\ B_r(G_n, U_n') \cong H_2\big) \to \mathbf{P}_{H_1}\mathbf{P}_{H_2}.$$

With $H_1 = H_2 = H$ this says that the second moment of $n^{-1}\#\{u : B_r(G_n, u) \cong H\}$ converges to $\mathbf{P}_H^2$, and the first moment converges to $\mathbf{P}_H$; hence the convergence in probability. Since balls with more than M vertices have expected proportion at most $(\sum_{l \le r} B^l)/M$, the convergence for each shape gives convergence in probability of the empirical average of every bounded function of the ball of radius r, which is local convergence in probability. The properties of the limit were proved in Corollary 7. □

5. Unbounded sublinear weights

We now assume only (E1)–(E3): $f \ge \alpha > 0$, $|f(k+1) - f(k)| \le L$ and $f(k)/k \to 0$. A bounded f with $\inf f > 0$ satisfies them, with $L = \beta - \alpha$. By (E3), for every $\varepsilon > 0$ there is

$a_\varepsilon \geq 1$ with

$$f(k) \leq \varepsilon\,(k + a_\varepsilon) \qquad \text{for all } k \geq 0. \tag{12}$$

The plan of the proof is the same, and we describe the changes section by section. The bounds $f \leq \beta$ and $C = \beta/\alpha$ were used in three places: the single-slot bound $C/(t-1)$ behind Section 2, the boundedness of the operator Q in Section 3, and the couplings of Section 4.

5.1. **Moments of a degree and the edge bound.** Let E be a multiset of pairs of labels with a slot assignment as in the proof of Lemma 3. The forced process, in which each assigned slot chooses its prescribed endpoint, and the identity (9) are defined as in Section 4; the identity holds for every positive f.

Lemma 20. *Let $\theta \in (0, 1/2)$ and let $p \geq 1$, $q \geq 0$ be integers. There are $a \geq 1$ and $K < \infty$ such that, in the original process and in every forced process with at most q assigned slots, for all $i < j$,*

$$\mathbb{E}^\sigma\big[(D_i + a)^p\big] \leq K\,(j/i)^{p\theta},$$

where D_i is the in-degree of i at any moment up to the last slot of vertex j.

Proof. Let $\varepsilon = \alpha\theta/(2m)$ and choose $a \geq a_\varepsilon$ with $(1 + 1/a)^{p-1} \leq 2$. Put $Z = D_i + a$. At a slot of a vertex $t + 1$ that is not assigned, i is chosen with conditional probability at most $f(D_i)/(\alpha t) \leq \varepsilon Z/(\alpha t)$, and $(Z + 1)^p - Z^p \leq p\,(1 + 1/a)^{p-1} Z^{p-1}$. Hence the conditional expectation of Z^p after the slot is at most $(1 + 2p\varepsilon/(\alpha t))Z^p$, and over the m slots of the vertex the factor is at most $\exp(2pm\varepsilon/(\alpha t)) = \exp(p\theta/t)$. An assigned slot multiplies Z^p by at most $(1 + 1/a)^p$. Since $\sum_{s=i}^{j-1} 1/s \leq 1 + \log(j/i)$, the claim holds with $K = a^p(1 + 1/a)^{pq} e^{p\theta}$. □

Lemma 21 (Joint edge bound)**.** *Let $\theta \in (0, 1/2)$. For every e there is K_e such that, for every multiset E of e pairs of distinct labels, the probability that G_n contains every pair of E with at least the prescribed multiplicity is at most*

$$K_e \prod_{\{i,j\} \in E} \max(i, j)^{\theta-1} \min(i, j)^{-\theta}.$$

Proof. Fix a slot assignment, and let ε, a and K be as in Lemma 20 and its proof for $p = q = e$, so that $f(k) \leq \varepsilon(k + a)$. By (9) with $\Psi = 1$, the bound $S^- \geq \alpha(j - 1) \geq \alpha j/2$, Hölder's inequality with exponent e, and Lemma 20, the probability that all assigned slots choose their prescribed endpoints is at most

$$\prod_{j \to i} \frac{2\varepsilon}{\alpha j} \Big(\mathbb{E}^\sigma\big[(D^-_{j\to i} + a)^e\big]\Big)^{1/e} \leq \prod_{j \to i} \frac{2\varepsilon K^{1/e}}{\alpha}\, j^{\theta-1} i^{-\theta},$$

where $j \to i$ runs over the pairs of E with $j > i$. There are at most m^e slot assignments. □

Let $H_\theta(i, j) = \max(i, j)^{\theta-1} \min(i, j)^{-\theta}$. With $w_i = i^{-1/2}$,

$$\sum_{j \leq i} H_\theta(i, j) w_j = i^{\theta-1} \sum_{j \leq i} j^{-\theta-1/2} \leq \frac{i^{-1/2}}{1/2 - \theta}, \qquad \sum_{j > i} H_\theta(i, j) w_j = i^{-\theta} \sum_{j > i} j^{\theta-3/2} \leq \frac{i^{-1/2}}{1/2 - \theta},$$

so the argument of Lemma 4 gives $\|H_\theta\| \leq 2/(1/2 - \theta)$. Take $\theta = 1/4$. The proofs of Section 2 then go through with Lemma 21 in place of Lemma 3: for each r there is K_r with

$$\mathbb{E}|B_r(G_n, U_n)| \leq K_r, \qquad \mathbb{P}\big(B_r(G_n, U_n) \cap [K] \neq \emptyset\big) \leq K_r \sqrt{K/n};$$

if all labels exceed εn, every factor in Lemma 21 is at most $1/(\varepsilon n)$, which is what the proof of Proposition 6 uses; and Corollary 7 is unchanged. Lemma 20 with $p = 2$, $q = 0$ gives

$\mathbb{E}D_i(n)^2 \le K(n/i)^{1/2}$, hence $\mathbb{E}D_{U_n}(n)^2 \le 2K$ and $\mathbb{P}(\max_i D_i(n) \ge \delta n) \le 2K/(\delta^2 n)$, which replaces Lemma 8.

5.2. **The normalisation.** The operator Q is no longer bounded on ℓ^1, and F is not Lipschitz. On the compact set $\mathcal{K}$, however, (12) gives for $y, z \in \mathcal{K}$ and $J \ge 1$

$$\begin{aligned} &\alpha \le s(y) \le \varepsilon(m + a_\varepsilon), \qquad \sum_{k \ge J} f(k)\, y_k \le m \sup_{k \ge J} \frac{f(k)}{k}, \\ &\sum_k f(k)|y_k - z_k| \le \max_{k<J} f(k)\, \|y - z\|_1 + 2m \sup_{k \ge J} \frac{f(k)}{k}. \end{aligned} \tag{13}$$

Since $\|Qy\|_1 \le 2\sum_k f(k)|y_k|$, the maps s, Q and F are uniformly continuous on $\mathcal{K}$ for the ℓ^1 norm, $\|F(y)\|_1 \le 2 + 2m$, and $\sup_{y \in \mathcal{K}} \sum_{k>J} |F(y)_k| \to 0$ as $J \to \infty$.

The recursion. Let $b_n = \max_{k \le mn} f(k)$, so that $b_n/n \to 0$. During the slots of vertex $n+1$ the count vector N changes by at most $2m$ in ℓ^1, so QN changes by at most $4mb_n$, and S changes by at most mL by (E2), while $S \ge \alpha n$ and $\|QN\|_1 \le 2S$. Hence (3) holds with the same η_{n+1} and with $\|\rho_n\|_1 \le K_0(b_n + 1)/n \to 0$. In the proof of Lemma 11 the ρ terms now contribute at most $(T+1)\sup_{n \ge N} \|\rho_n\|_1$, and the F terms are controlled by the uniform tail bound above; the conclusion of the lemma holds.

The semigroup. The pure birth process X with rates $f(k)$ does not explode, because by (12) $\mathbb{E}_k(X_t + a_\varepsilon) \le (k + a_\varepsilon)e^{\varepsilon t}$. Its semigroup P_t acts on ℓ^1 as a strongly continuous semigroup of positive contractions that preserve mass, and it is bounded and strongly continuous for the norm $\|y\|_{(1)} = \sum_k (1+k)|y_k|$. The operator Q is bounded from this weighted space to ℓ^1. For a finitely supported y, $P_t y - y = \int_0^t P_u Q y \,\mathrm{d}u$; by density this extends to all y with $\|y\|_{(1)} < \infty$, so such y belong to the domain of the generator of (P_t) on ℓ^1, and the generator acts on them as Q. For $y \in \mathcal{K}$ put $g_y(t) = s(P_t y)$ and $h(t) = s(P_t e_0)$; then, for every $\varepsilon > 0$,

$$\alpha \le g_y(t) \le \varepsilon(m + a_\varepsilon)e^{\varepsilon t}, \qquad \alpha \le h(t) \le \varepsilon a_\varepsilon e^{\varepsilon t}, \qquad \mathbb{E}_0 X_t = \int_0^t h(u)\,\mathrm{d}u. \tag{14}$$

Lemma 22. *Let $y : [0,T] \to \mathcal{K}$ be a solution of $y(\tau) = y(0) + \int_0^\tau F(y(u))\,\mathrm{d}u$. Define the clock $t(\tau)$ by $\mathrm{d}t/\mathrm{d}\tau = m/s(y(\tau))$, $t(0) = 0$, let $z(t) = y(\tau(t))$ and $w(t) = z(t)\exp\bigl(\frac{1}{m}\int_0^t s(z(u))\,\mathrm{d}u\bigr)$. Then w and $\sigma = s(w)$ satisfy (5) with initial vector $y(0)$. In particular y is determined by $y(0)$.*

Proof. The integrand is continuous, so y is continuously differentiable in ℓ^1, and so are z and w. As in Lemma 12, $w' = Qw + \frac{1}{m} e_0\, s(w)$, where $w(t)$ has finite mean and therefore lies in the domain of the generator. For fixed t the function $u \mapsto P_{t-u}w(u)$ is then differentiable on $[0,t]$ with derivative $P_{t-u}(w'(u) - Qw(u)) = \frac{1}{m} P_{t-u} e_0\, \sigma(u)$, which gives the first equation of (5). Applying the positive linear functional s to it gives the second. The functions g_y and h are locally bounded by (14) and σ is continuous, so σ is the unique locally bounded solution of this Volterra equation; it determines w, then $z = w/|w|$, then the clock. □

Uniform attraction. By (14) the Laplace transform of h is finite on $(0,\infty)$, continuous and strictly decreasing, at least α/γ at γ, and tends to 0 at infinity. Let $\gamma > 0$ be the point where it equals m and $\kappa(t) = e^{-\gamma t}h(t)/m$. Taking $\varepsilon < \gamma$ in (14), κ is continuous and bounded by a multiple of $e^{-(\gamma-\varepsilon)t}$, hence directly Riemann integrable, and $\tilde{g}_y(u) = e^{-\gamma u} g_y(u)$ lies between $\alpha e^{-\gamma u}$ and $\varepsilon(m + a_\varepsilon)e^{-(\gamma-\varepsilon)u}$ for all $y \in \mathcal{K}$. The proof of Proposition 13 now applies word for word to the functions w of Lemma 22: $z(t) \to \pi = \gamma \int_0^\infty e^{-\gamma u} P_u e_0 \,\mathrm{d}u$ in ℓ^1 as $t \to \infty$, uniformly in $y(0) \in \mathcal{K}$, and the clock satisfies $t \ge m\tau/(\varepsilon(m + a_\varepsilon))$. The vector π is $p(m\gamma)$; by

the last identity of (14) its mean is $\int_0^\infty e^{-\gamma u}h(u)\,\mathrm{d}u = m$, and $s(\pi) = \gamma\int_0^\infty e^{-\gamma u}h(u)\,\mathrm{d}u = m\gamma$. The function M of Section 3 is finite, since $M(d) = \int_0^\infty e^{-t}\,\mathbb{E}_0 X_{mt/d}\,\mathrm{d}t$ and $\mathbb{E}_0 X_t \le a_\varepsilon e^{\varepsilon t}$ for every $\varepsilon > 0$, and it is strictly decreasing, so $c_f = m\gamma$ is the only solution of $M(d) = m$. This proves Lemma 9(iii) in the present setting, without the bounds $\alpha \le c_f \le \beta$; one has $c_f \ge \alpha$.

Proof of Theorem 10. Suppose that, with positive probability, $\|y(\tau_k) - \pi\|_1 > 2\delta$ along a sequence $\tau_k \to \infty$. Fix T such that every solution $y : [0,T] \to \mathcal{K}$ of the integral equation of Lemma 22 has $\|y(T) - \pi\|_1 \le \delta$. The paths $u \mapsto y(\tau_k - T + u)$, $u \in [0,T]$, take values in the compact set $\mathcal{K}$ and are uniformly Lipschitz, so a subsequence converges uniformly to some $\bar y$. By (3), Lemma 11, $\|\rho_n\|_1 \to 0$ and the uniform continuity of F on $\mathcal{K}$, the limit satisfies $\bar y(u) = \bar y(0) + \int_0^u F(\bar y(v))\,\mathrm{d}v$. Hence $\|\bar y(T) - \pi\|_1 \le \delta$, a contradiction. Thus $y_n \to \pi = p(c_f)$ almost surely in ℓ^1, and $S_n/n = s(y_n) \to s(\pi) = c_f$ because s is continuous on $\mathcal{K}$.

5.3. **The local limit.** Since s is bounded on $\mathcal{K}$, (6) holds. In Lemma 17 replace $\bar K$ by a bound for the ℓ^1 distance $\bar d$ between the two in-degree sequences: by (E2) the two weight vectors differ by at most $L\bar d$ in ℓ^1, the two choices at a slot of vertex $t+1$ differ with probability at most $L\bar d/(\alpha t)$, a disagreement or an assigned slot increases $\bar d$ by at most 2, and the two values of S_t differ by at most $L\bar d$. This gives $\mathbb{E}\bar d \le K_\varepsilon$ and the conclusion of the lemma.

In Lemma 18, fix M and stop the comparison when a tagged in-degree exceeds M. Before that time only the values $f(0),\dots,f(M)$ are used and the proof applies. By Lemma 20 with $p = 1$ the probability that a tagged in-degree exceeds M before time n is at most K_ε/M, uniformly in n and in the labelling, and the processes Y_i do not explode. Letting $n \to \infty$ and then $M \to \infty$ gives the convergence in total variation.

In Proposition 19 the product of the numerators $f(D^-_{j\to i})$ is not bounded, but by Hölder's inequality and Lemma 20 with $p = 2e$ its second moment under $\mathbb{P}^\sigma$ is bounded uniformly in n and in the labellings with labels at least εn. The factors n/S^- are at most $2/(\alpha\varepsilon)$ and converge in probability. Uniform integrability therefore gives the same limit and a uniform bound for $n^e\mathbb{P}(E_{\iota,\sigma} \cap D_\iota)$.

The proof of Theorem 15 then applies with Lemma 21 in the estimate of the extra edge and the ball bounds of this section in the removal of the cutoff and in the two-root step. Proposition 16 holds as stated. This proves Theorem 1.

6. Simulations

The process was simulated as defined, with $m = 2$ and $n = 10^6$ vertices, for the two-level rule $f(0) = 1$, $f(k) = 2$ $(k \ge 1)$ of Example 14 and for $f(k) = \sqrt{k+1}$. For the two-level rule with $f(0) = a$ and $f(k) = b$ one has $h_\lambda(u) = b + (a-b)e^{-\lambda a u}$, so by Remark 2 the ratio of the age of a parent to the age of its child has density $(b + (a-b)v^{\lambda a})/c_f$ on $(0,1)$, with mean $(b/2 + (a-b)/(\lambda a + 2))/c_f$. Let H be the ball of radius 2 in which the root has no child and each of its m parents has no child other than the root. In Proposition 16 the root, of age x, contributes $x^{\lambda a}$, a parent of age y contributes $a(y/x)^{\lambda a}x^{\lambda b}/(c_f x)$, each grandparent integrates to one, and the slot factors cancel the automorphisms, so

$$\mathbf{P}_H = \int_0^1 x^{\lambda a}\Big(\frac{a\,x^{\lambda b}}{c_f(1+\lambda a)}\Big)^m \mathrm{d}x = \frac{(a\,p_0/c_f)^m}{1+\lambda a + m\lambda b}, \qquad p_0 = p_0(c_f) = \frac{1}{1+\lambda a}.$$

Table 1 gives the limits and the values observed in four independent runs. The age ratio is measured for the first edge of the vertices $n/2,\dots,n$, the other quantities over all vertices.

A proportion p measured on 10^6 vertices has standard deviation about $\sqrt{p(1-p)} \times 10^{-3}$ if the vertices are treated as independent, which is 5×10^{-4} for p_0 and 10^{-4} for $\mathbf{P}_H$. The differences in the table are of this order; the largest is about 1.8 such standard deviations.

	limit	run 1	run 2	run 3	run 4
$f(0) = 1$, $f(k) = 2$ for $k \geq 1$					
p_0	0.43845	0.43818	0.43821	0.43837	0.43879
p_1	0.15767	0.15773	0.15802	0.15727	0.15758
p_2	0.11340	0.11372	0.11323	0.11365	0.11301
p_3	0.08156	0.08158	0.08151	0.08178	0.08162
mean age ratio, parent to child	0.44519	0.44485	0.44540	0.44519	0.44539
$\mathbf{P}_H$ for the ball above	0.010648	0.010782	0.010718	0.010594	0.010709
$f(k) = \sqrt{k+1}$					
S_n/n	1.56616	1.56618	1.56619	1.56615	1.56637
p_0	0.43917	0.43919	0.43879	0.43960	0.43878
p_1	0.19987	0.19937	0.20022	0.19941	0.19973
p_2	0.11238	0.11266	0.11264	0.11224	0.11294
p_3	0.06994	0.07006	0.06971	0.07025	0.07005

TABLE 1. Limits from Theorem 1 and simulated values, $m = 2$, $n = 10^6$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [23] and Math Brain [21]. In October 2026 an AI agent connected to the Brain, working from the project's list of open problems in probability, obtained Theorem 1 and a proof; a Codex agent (OpenAI) and a Claude agent (Anthropic) each checked that argument, and the second wrote the simulation program. In November 2026 a Claude Fable agent (Anthropic) checked the proof again, ran the simulations and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/sublinear-attachment-local-limit`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [21].

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